Object Oriented Programming [ECEg-3102]

Chapter One:

Introduction to the OOP paradigm

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Outline

- Comparison between procedural/ structured programming and OOP.
- Basic concepts: Abstraction, Encapsulation or information hiding, Inheritance and Polymorphism.
- Program reusability and extensibility.

Comparison between procedural/ structured programming and OOP

There arc two common programming approach:

- Procedural programming and
- Object-oriented programming (or OOP)

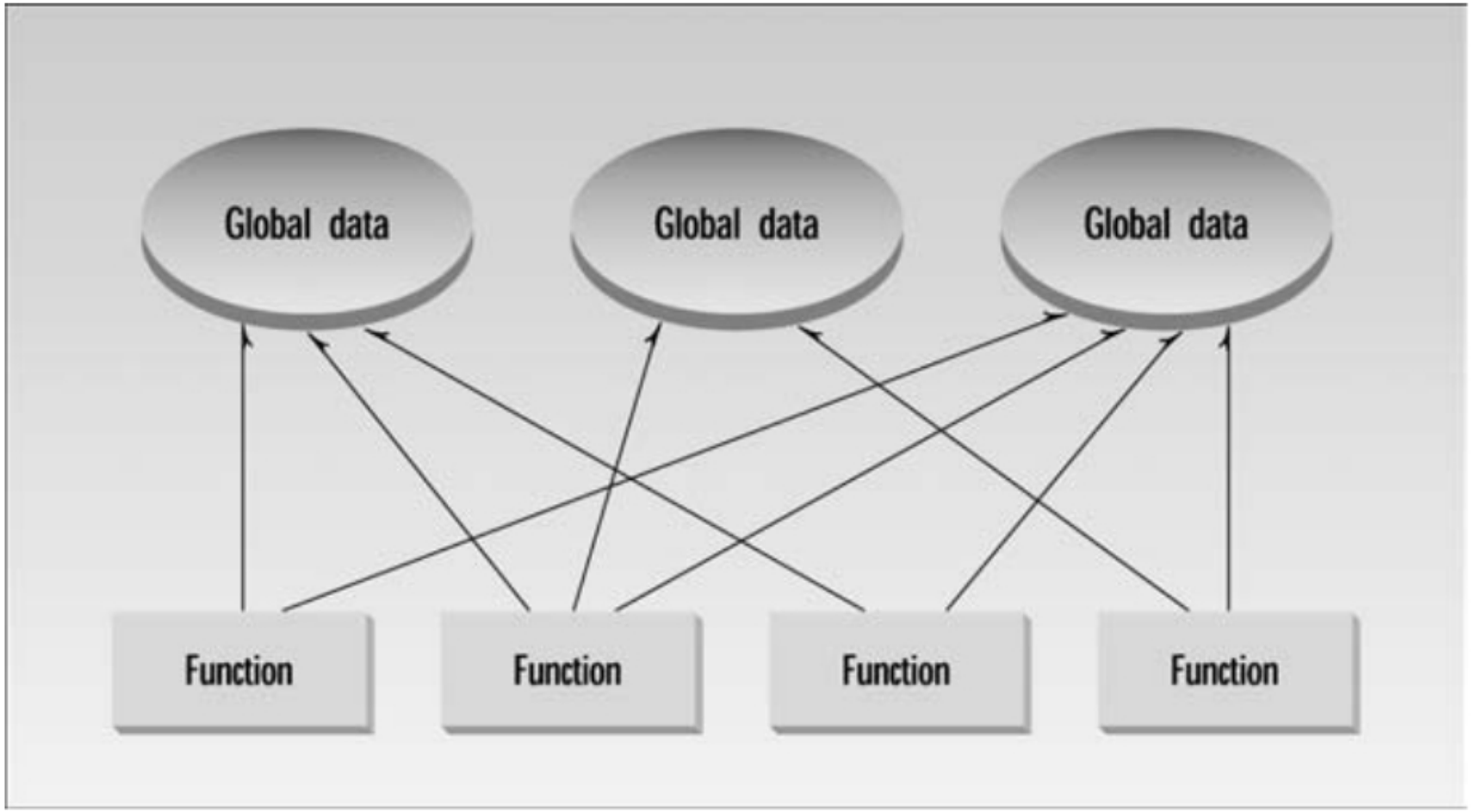
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- **Procedural programming** is a programming practice centered on the **procedures** or **actions** that take place in a program.
- **Object-oriented programming** is centered around the object.
  - **Objects** are created from abstract data types that encapsulate **data** and **functions** together.

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- In procedural programming the **data** and the **functions** are separate entities.
- When programs become larger and more complex, the separation of a program's data and the code that operates on the data can lead to problems.

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The procedural paradigm.

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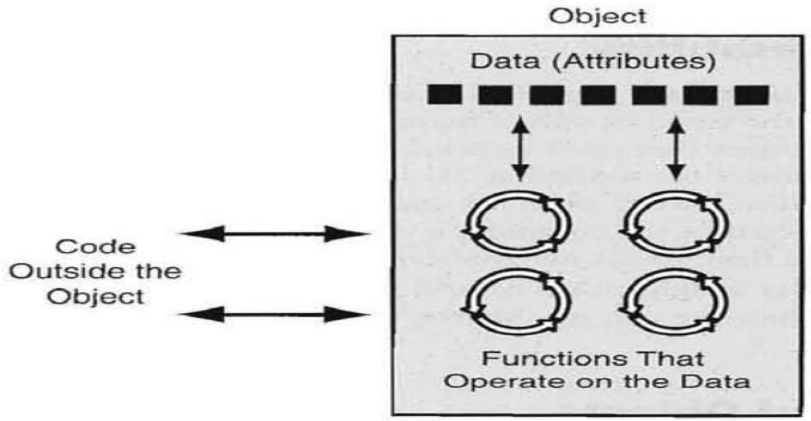
- When the structure of the **data changes**, the **code that operates** on the data must also change to accept the new format.
  - This results in additional work for programmers and a greater opportunity for bugs to appear in the code.
  - This problem has helped influence the shift from procedural programming to object-oriented programming.

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- An **object** is a software entity that contains both data and procedures.
- The object is, conceptually, a self-contained unit consisting of attributes (data) and procedures (functions).

Cont'd...

- **OOP** addresses the problems that can result from the separation of code and data through **encapsulation** and **data hiding**.
- **Encapsulation** refers to the combining of data and code into a single object.
- **Data hiding** refers to an object's ability to hide its data from code that is outside the object.



Basic concepts

- Abstraction
- Encapsulation or information hiding
- Inheritance and
- Polymorphism

Abstraction

- **Abstraction** refers to the act of representing essential features without including the background details or explanations.

  Example We can driving a car with out knowing how each system work

- We can **use** object without knowing its internal **implementation**.

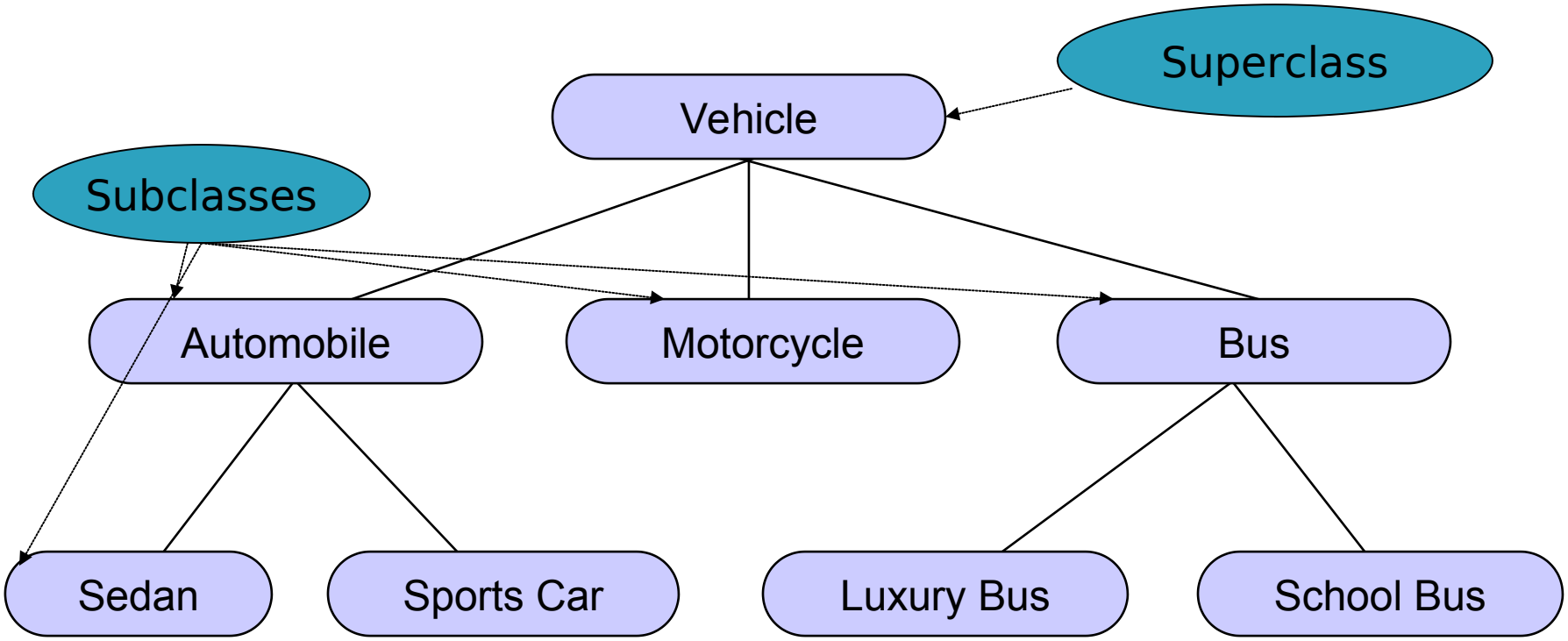
Encapsulation

- **Encapsulation:** wrapping up of data and methods into a single unit (called class)
- **Data** is not accessible to the outside world and only those methods in the class can access it.
- Insulation of the data from direct access by a program is called data hiding.

Inheritance

- **Inheritance** is a process by which object of one class acquire the property of object of other class.
- Classes with properties in common can be grouped so that their common properties are only defined once.
  - **Superclass** – inherit its attributes & methods to the subclass(es).
  - **Subclass** – can inherit all its superclass attributes & methods besides having its own unique attributes & methods.

Cont'd...


Superclass
Vehicle
Subclasses
Automobile
Motorcycle
Bus
Sedan
Sports Car
Luxury Bus
School Bus

Polymorphism

- The ability to take more than one form
- The same word or phrase can mean different things in different contexts

Analogy:

- In English, bank can mean side of a river or a place to put money.

Program reusability and extensibility

- Re-usability – without redundant code
  - Sharing a certain code within a program
  - Previously written code can be used again in the new code.
- Extensible – possible to alter additional features to the existing class.

Thank You !